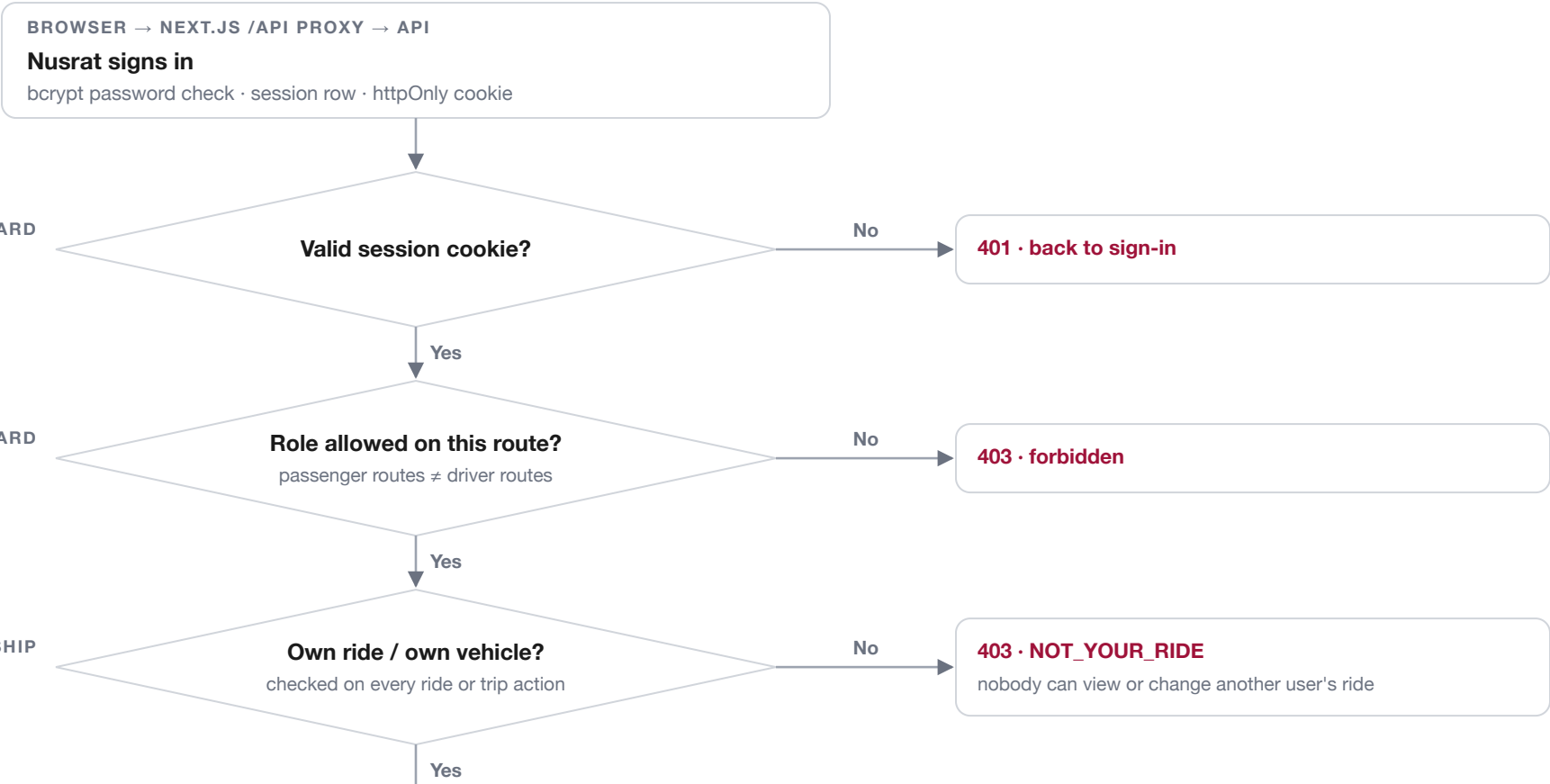


Dhaka Tesla Pool · Ride Flow

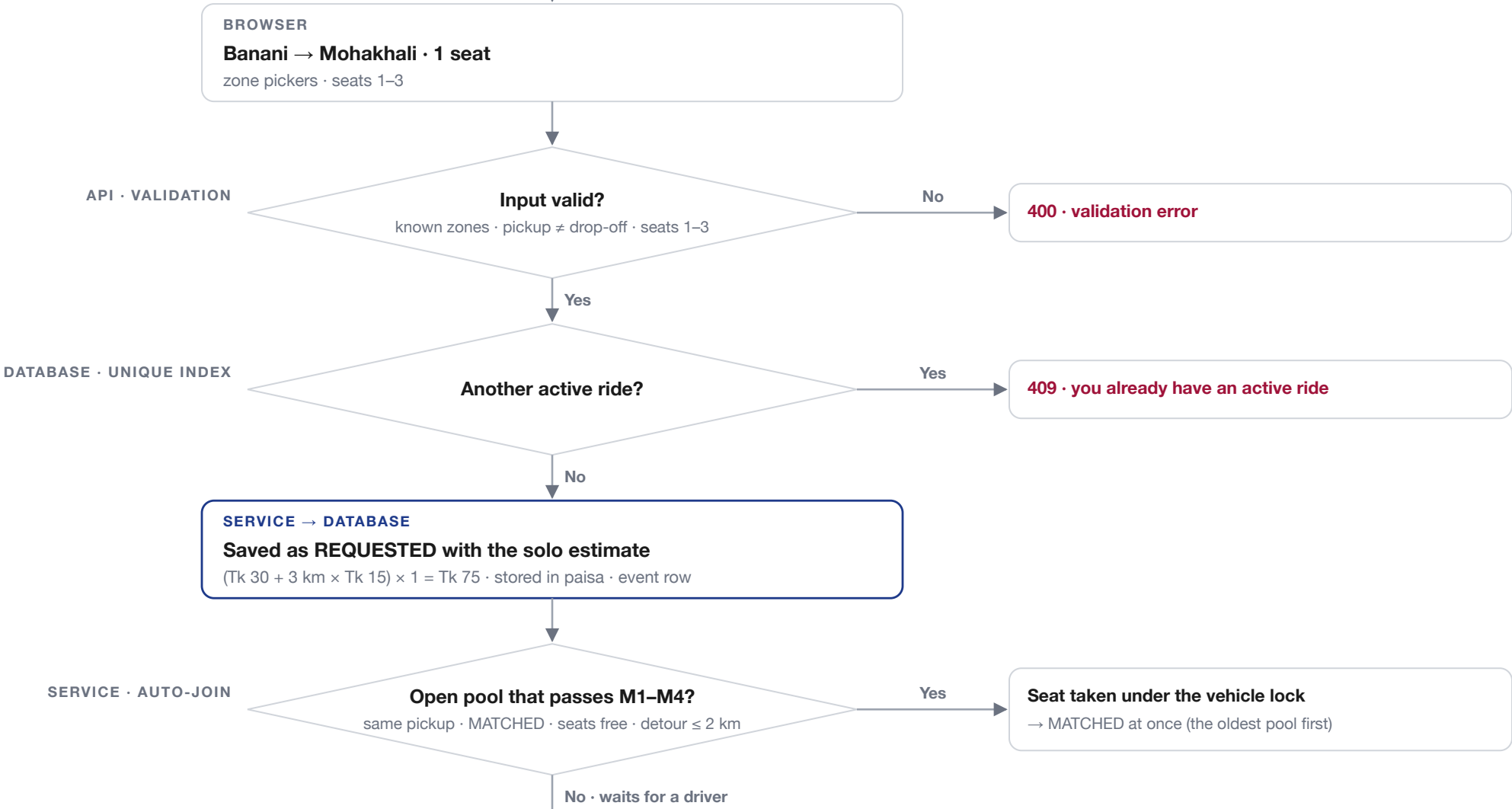
How one ride moves through the system · read top to bottom · side notes cover concurrent and edge cases

1 SIGN IN



Sign up
Passengers only; drivers are seeded
Unique email (stored lower case)
Password hashed with bcrypt
Login rate-limited: 5 per minute

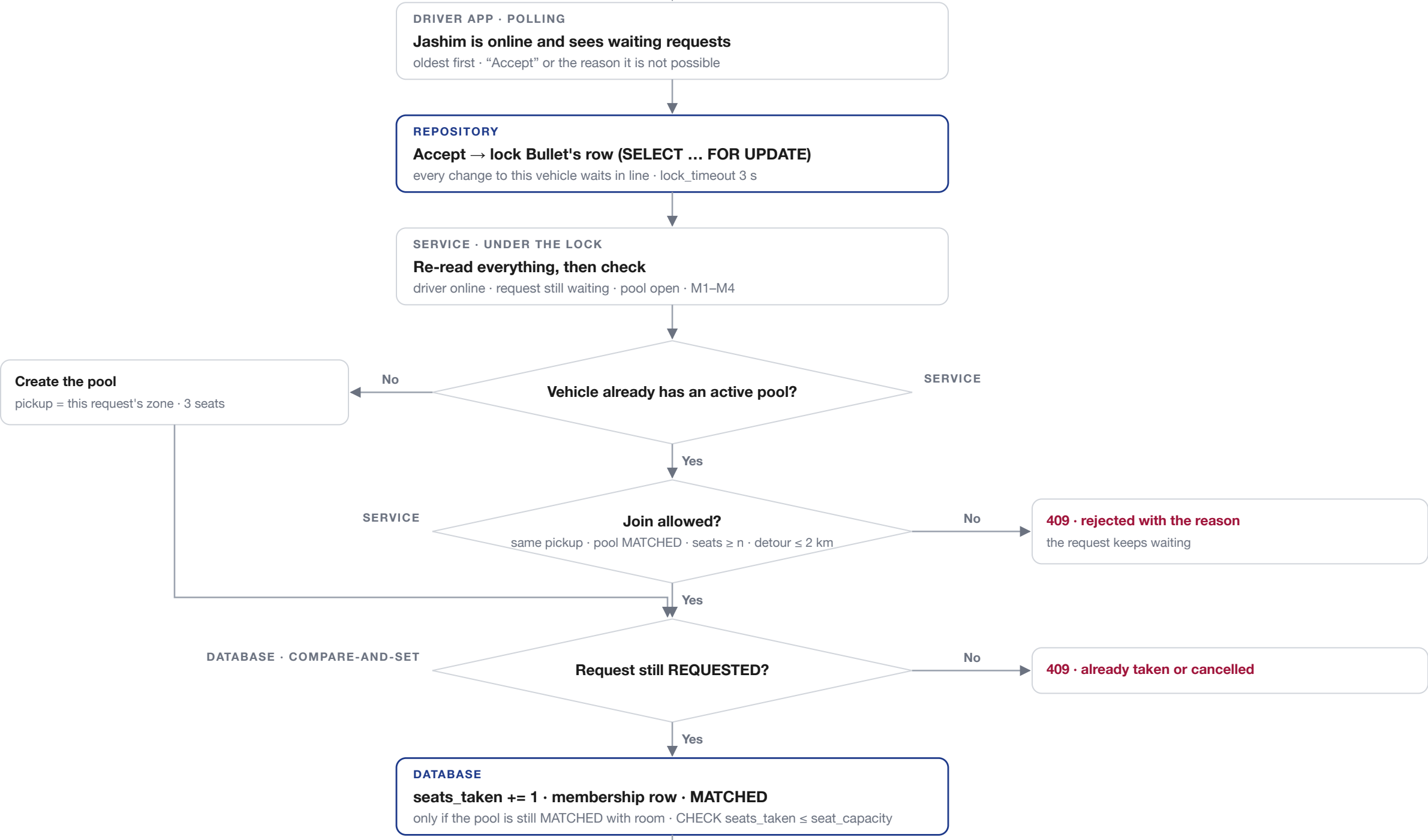
2 REQUEST A RIDE



Fare model
(Tk 30 + km × Tk 15) × seats
–20% if 2+ passengers at start
Estimate = solo fare, never exceeded
Discount counts passengers, not seats

Detour rule (M4)
Drop-offs nearest first
detour = in-car km – direct km
Rafiq via Mohakhali: 6 – 4 = 2 km ✓
Banani → Uttara: 16 – 12 = 4 km ✗

3 MATCHING

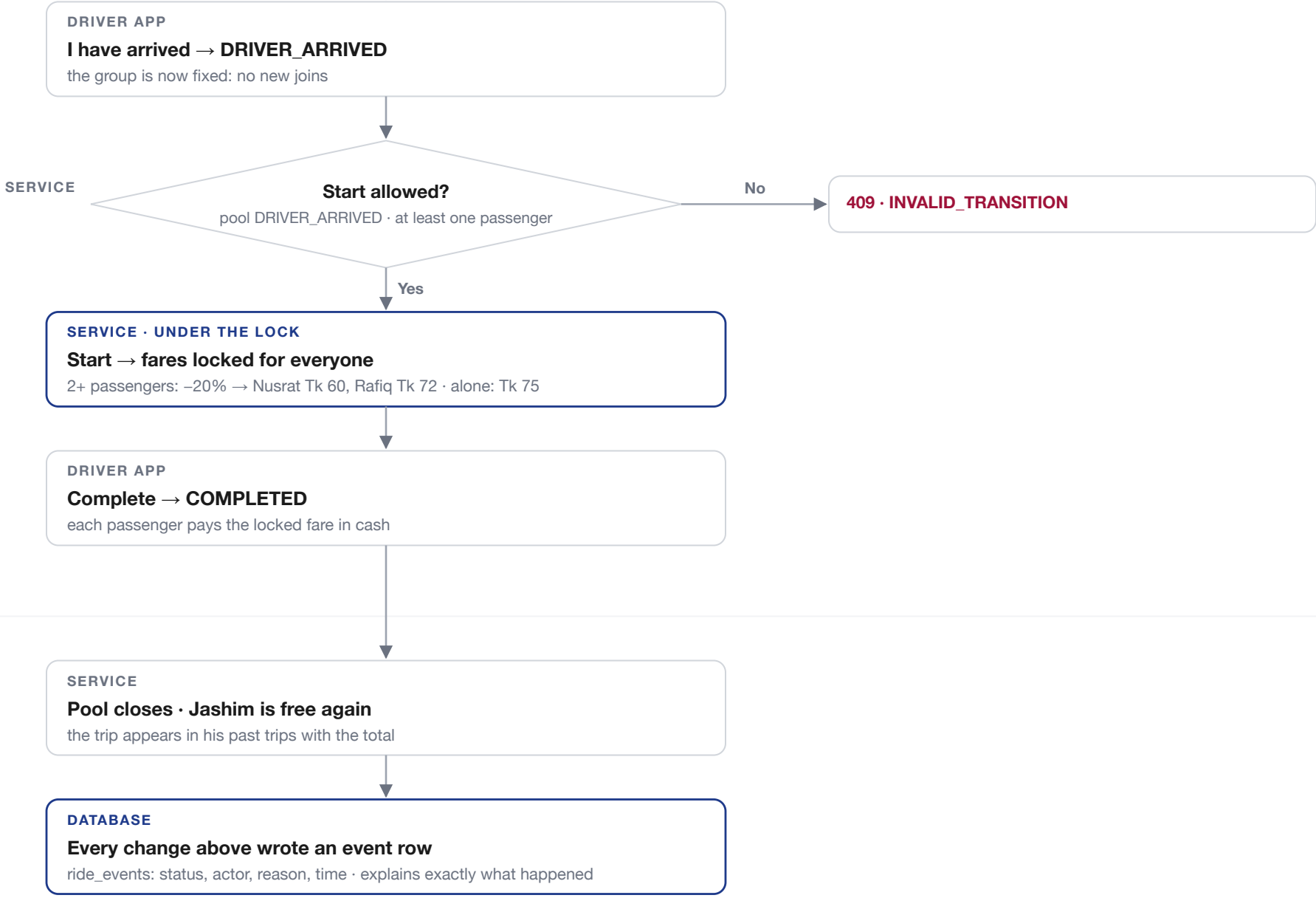


Last seat, many riders at once
Nusrat holds 2 of Bullet's 3 seats
8 riders ask for the last one
All wait for Bullet's lock
Exactly 1 wins · 7 keep waiting

No pool yet, two accepts at once
First accept creates the pool
Second waits, then joins it
One pool, two passengers

Retries and timeouts
Lock wait > 3 s → 503 BUSY
Auto-join is best effort: if busy, the ride just keeps waiting

4 THE TRIP



Online / offline
Changed under the vehicle lock
Offline refused during a trip
Accept refused while offline

What the passenger sees
Polls her ride every 3 seconds
Own status and own fare only
Driver + vehicle name
Co-riders' first names only

Cancellation rules
Passenger: before the start
→ seats freed at once
Last one out → pool closes
Driver: before the start → passengers back to REQUESTED
After the start: no cancel (409)

5 END OF RIDE